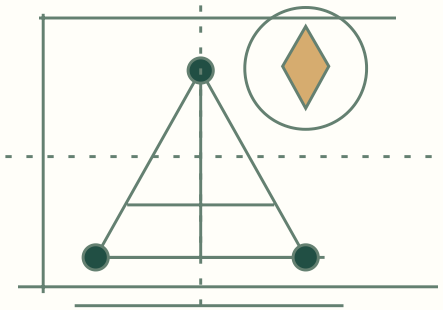
# Whitney Learning Compass



## From Curriculum Goals to Hands-On Discovery

An evidence-grounded AI decision assistant connecting teachers' learning objectives to real museum workshops and editable learning journeys.

**Anthony Qiao**

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Joint MBA/MEM Candidate

Independent academic prototype. Not affiliated with or endorsed by the museum.

# The decision goes beyond finding a workshop

For educators, initially Grades 2–7, a workshop must fit both a learning objective and the practical limits of a class visit.

01

## Find a workshop

Explore hands-on science, engineering and design.

02

## Judge the fit

Check grades, program fees and time. Keep unresolved logistics visible.

03

## Plan the learning

Connect preparation, workshop inquiry and classroom follow-up.

**Eli Whitney Museum & Workshop**

A nonprofit in Hamden, Connecticut, centered on making and experimentation. Its website already provides useful program filters.

**A personal starting point**

My Golub Capital Board Fellows Program match inspired this independent project. The planning problem began as a design hypothesis, not validated educator demand.

Final report pp. 2–3 · Public museum information · No museum commission or endorsement.

# One connected curriculum-to-workshop workflow

Source-grounded selection comes before instructional generation.

**01** Learning objective
Teacher goals and class constraints

→

**02** Verified matching
Grades, published fees and duration

→

**03** AI recommendations
Educational reasoning for up to three real choices

→

**04** Teacher selection
An explicit human choice before the next agent runs

→

**05** Editable learning journey
Before / During / After, review and printing

**20**
real programs in the curated catalog

**20**
published per-student fees

**18**
published durations recorded

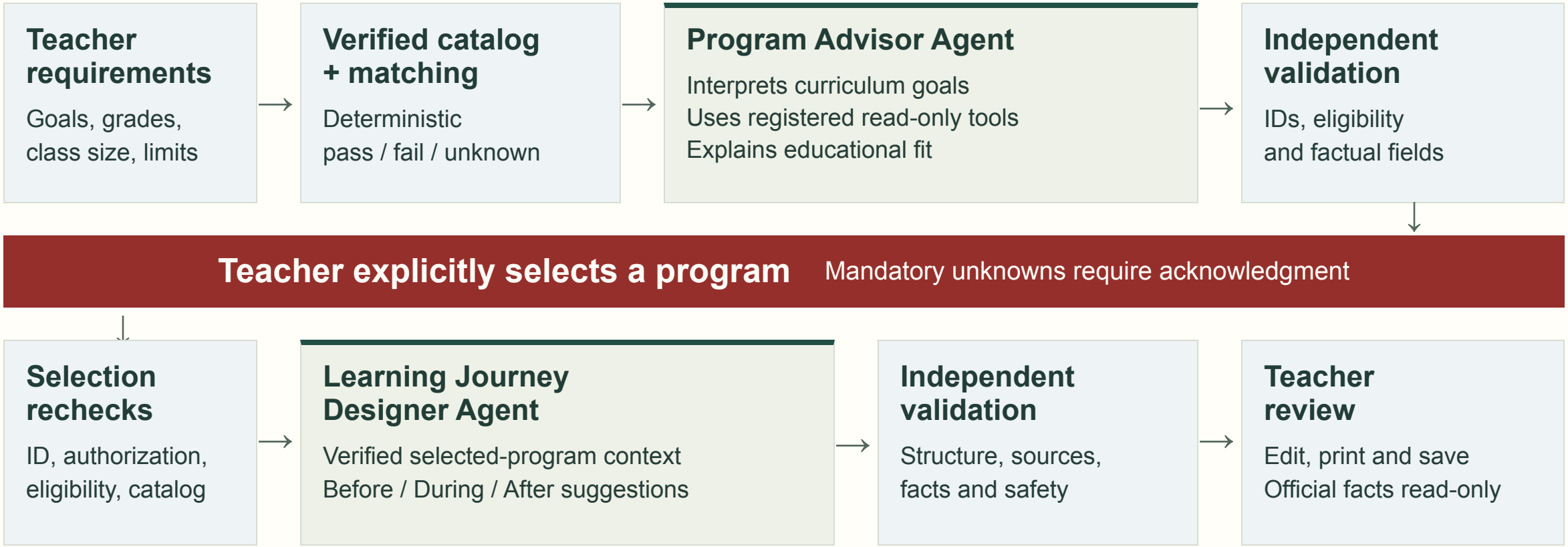
The contribution is the integration: educational intent, traceable facts, deterministic checks and teacher control. Local draft recovery and an editable museum inquiry support follow-through.

Thematic coverage is incomplete. Missing facts remain unknown. USD is a disclosed application assumption.

AGENT DESIGN

# Two specialists. One teacher in control.

**FastAPI coordinates the sequence.** Both PydanticAI agents use **gpt-6-luna** through Portkey.

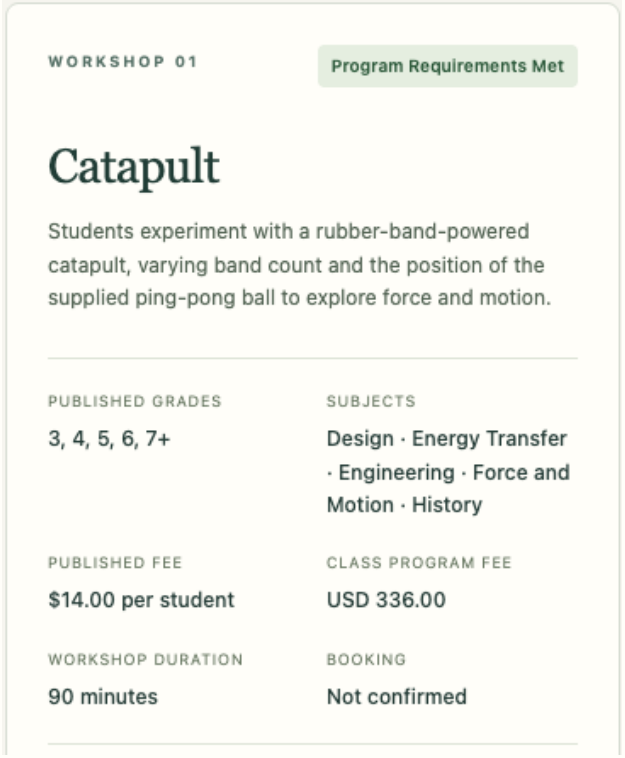



**AI:** educational reasoning **Code:** authoritative checks **Human:** choice and judgment

# A teacher's goal becomes a validated choice

**Grade 4 · 24 students · $400 program-fee budget · 90-minute maximum**

Learning objective: force, motion and engineering design. Explicit USD assumption.




Unaltered crop of the live recommendation.

## Advisor's educational reasoning

WHY THIS COULD FIT · AI INTERPRETATION

The catapult activity directly connects hands-on experiments with force and motion: students vary the setup and can compare how those changes affect the ball's movement.

Capacity, availability, accessibility, delivery, extra charges and booking conditions require museum confirmation.

**RECORDED TOOL TRACE · SUMMARY**

`search_matching_programs`

16 catalog candidates
14 pass · 2 unresolved
Then `final_result`
**3 validated recommendations**

## Recorded class program fees

| | |
|---|---|
| **Catapult** | **$336** |
| Marble Roller Coaster | **$336** |
| The Whitney Relay | **$240** |

## Teacher explicitly selects Catapult




Recorded review before the Designer request.

# An editable plan for the selected Catapult workshop

**Before**

## Prepare & Wonder

Prepare a simple paper-and-pencil recording sheet with columns for trial or setup, prediction, actual observation, and possible explanation; leave observation fields blank until observations occur.

**During**

## Explore & Observe

**Evidence collection**

Record the setup and the directly observed result in words or a simple sketch; do not fill in results before observing them.

**After**

## Reflect & Extend

**Learning assessment**

Objective 1: Review each student's explanation for a connection between a changed setup feature (band count or ball position) and motion, supported by an actual observation. If none is available, assess the student's ability to state what evidence would be needed instead.

☑ I have reviewed these teaching suggestions, safety notes and unresolved requirements.

Print / Save as PDF

**RECORDED DESIGNER TRACE · SUMMARY**

`get_selected_program`

Verified Catapult context

Then `final_result`

**Three validated phases**

## The teacher stays in control

Edit suggestions, review the plan, then print or save a copy.

This live workflow included an objective edit and a five-page printout. The museum inquiry was edited and copied locally, never sent.

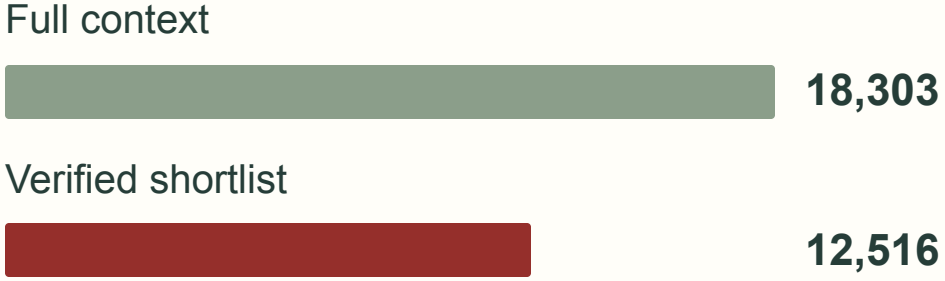
Local draft recovery was tested separately offline. It preserves editable work, not current museum verification or backend authorization.

TOKENOMICS

# Less Advisor context, lower illustrative token cost

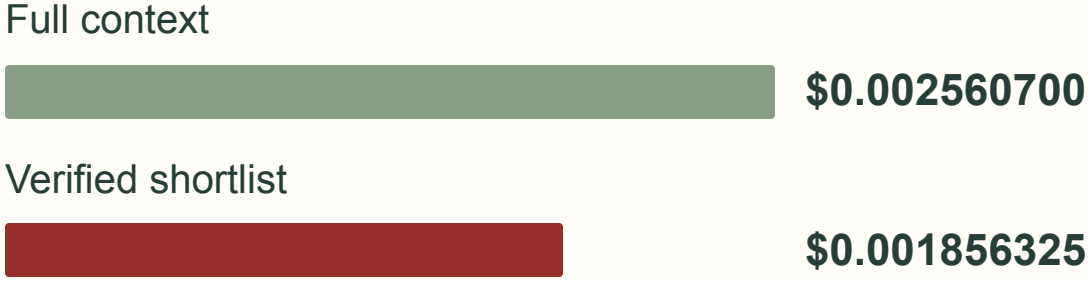
## Advisor input tokens

**31.62%** reduction




## Illustrative Advisor cost

**27.51%** reduction




Three paired Advisor tasks · Six successful runs · Bars start at zero on a common linear scale within each chart.

### SEPARATE SUCCESSFUL HISTORICAL TWO-AGENT WORKFLOW

| 9,708 in / 1,242 out | 4 requests | $0.0015918–$0.0018345 |
|---|---|---|
| Actual provider-reported tokens | Two per agent | Public-rate estimate, cache writes unknown |

**Advisor comparison only.** Output quality was not proven equivalent. The paired Designer comparison was not completed. Public-rate estimates, not verified gateway billing. Hosting, engineering, support, retries and other overheads excluded.

# Verified constraints outrank AI reasoning

**OFFLINE ADVERSARIAL TEST**

**$335**
Teacher's budget

**$336**
Verified Catapult program-fee subtotal

Simulated model proposal: **eligible**
Deterministic eligibility: **fail**
Independent validator: **rejected**

Actual application validation logic.
No live-model attack was performed.

**24/24** controlled technical scenarios

**33/33** specified eligibility decisions

## Wrong fact accepted

The Advisor called Catapult's supplied ping-pong ball a marble. A targeted material guard now rejects the preserved output.

## Correct warning rejected

The Designer warned against invented observations. The guard misread the prohibition. Clause-aware validation now accepts the preserved text.

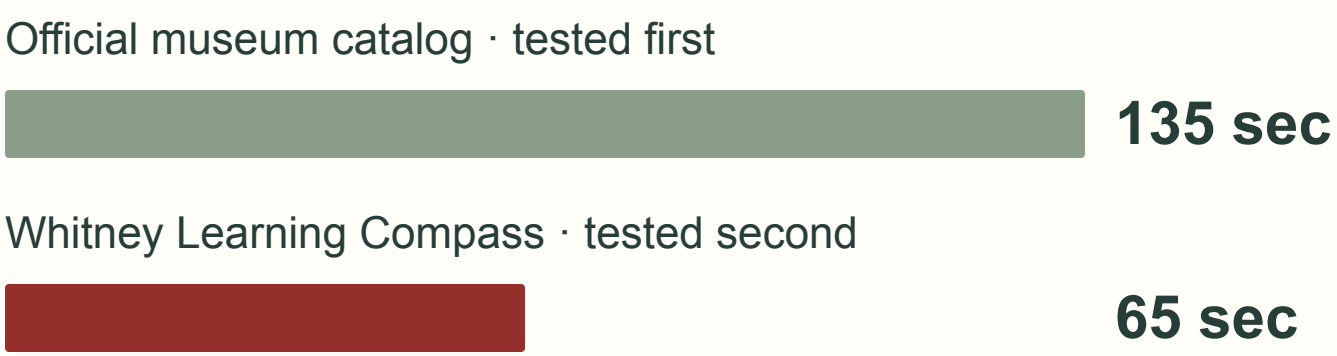
**Both repairs were verified offline.** Neither result proves general factual accuracy. Natural-language validation still misses or overblocks some claims.

Read-only tools, selection authorization and token-free local drafts protect other parts of the decision path.

EXPLORATORY IMPACT

# A faster selection in one proxy-user task

MEASURED ACTIVE PROGRAM-SELECTION TIME




## Participant ratings

| | |
|---|---|
| Website discovery | 3/5 |
| Compass discovery & comparison | 4/5 |
| Confidence in requirements | 4/5 |
| Learning-plan usefulness | 3/5 |
| Willingness to use for planning | 4/5 |

Integrated information helped comparison. The learning plan still felt generic, and intake needed more guidance.

**51.9%**

less measured active selection time

70 seconds in this single task

**One Yale SOM graduate-student proxy, not a K–12 teacher.** Website first, Compass second. Timing excluded AI waiting and later plan review. The result is exploratory, not causal or generalizable.

Final report pp. 11–12 · Actual session, Oct. 23, 2026 · Ratings are single-participant perceptions on a 1–5 scale.

# Credibility comes from the decision process

## A credible AI planning assistant is more than a good answer.

## Evidence is inspectable. Uncertainty is actionable. Human decisions are explicit.

An independent academic prototype.
No museum endorsement, deployment or measured classroom learning impact.

01 **Test with actual educators**
Include K–12 teachers and museum stakeholders. Measure end-to-end usefulness.

02 **Guide curriculum input**
Offer selectable subjects and topics alongside free-text goals.

03 **Make instruction more specific**
Customize the detail and activities in the learning journey.

04 **Broaden and refresh the evidence**
Expand source coverage and strengthen factual and safety checks.

05 **Assess operational feasibility**
Verify full-workflow cost, deployment needs and organizational readiness.

Final report p. 13 · These are proposed next steps, not implemented features or museum-approved plans.